Autonomous Semantic Mapping on Edge Devices: A Survey of VLA and Predictive NBV

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***Abstract*—Autonomous exploration of complex infrastructures demands the seamless integration of high-fidelity spatial perception with cognitive reasoning capabilities. This survey systematically reviews the convergence of three rapidly evolving research threads: (i) visual Simultaneous Localisation and Mapping (ViSLAM) augmented with 3D Gaussian Splatting (3DGS) for real-time, photorealistic scene reconstruction; (ii) Vision-Language-Action (VLA) foundation models that endow robots with grounded spatial understanding and natural-language instruction following; and (iii) predictive Next-Best-View (Pred-NBV) planners that minimise exploration entropy under energy-constrained budgets. We articulate how VLA-driven semantic intelligence can proactively guide metric reconstruction through closed-loop Pred-NBV strategies, reducing flight effort while maximising surface coverage. Moreover, we analyse the feasibility of deploying these architectures on edge-computing hardware (e.g., NVIDIA Jetson) via parameter-efficient fine-tuning techniques such as Low-Rank Adaptation (LoRA) and 8-bit quantisation. A comparative analysis of reconstruction quality, exploration efficiency, cognitive performance, and hardware utilisation metrics consolidates the state of the art and identifies open challenges toward fully autonomous, semantically aware mobile robots and unmanned aerial vehicles (UAVs).**

***Index Terms*—survey, visual SLAM, 3D Gaussian Splatting, vision-language-action models, next-best-view planning, edge computing, autonomous exploration, UAV**

I. Introduction

Autonomous inspection of built environments requires robotic platforms capable of constructing dense, geometrically faithful maps while reasoning about *what* to observe next and *why*. Traditional pipelines decouple spatial perception from high-level cognition: a SLAM front-end densifies the map, and a separate planner selects viewpoints purely on the basis of geometric information gain [1]. This decoupling introduces two critical limitations: the planner remains “blind” to semantic context, and the computational cost of both dense reconstruction and large vision-language models has historically confined these systems to cloud-based deployments, precluding real-time autonomy on UAVs.

Recent advances motivate a re-examination of this architecture. 3D Gaussian Splatting (3DGS) surpasses NeRF in efficiency for robotic applications [2], [3]. Vision-Language-Action (VLA) models—such as those built on Qwen-VL [4] and Lumo-1 [5]—interpret complex instructions with spatial awareness, achieving success rates above 90% in task switching [6]. Predictive NBV algorithms observe up to 25.46% more surface points within the same budget [1], [7].

This paper fills the gap of a unified survey by:

1) Providing a taxonomy of modern VLA models and their coupling with real-time 3DGS-based SLAM systems (e.g., MCGS-SLAM [8], MVS-GS [9]).
2) Analysing computational feasibility on embedded hardware through LoRA and quantisation [10].
3) Proposing a closed-loop architecture unifying dense perception (3DGS), semantic cognition (VLA), and active exploration (Pred-NBV / GenNBV).

II. Background and Related Work

A. Visual SLAM and 3D Reconstruction

Visual SLAM constitutes the perceptual backbone of autonomous mobile robots. Multi-camera frameworks such as MCGS-SLAM [8] fuse observations from multiple viewpoints into a unified Gaussian map, while MVS-GS [9] combines online multi-view stereo with Gaussian-based scene representations. Key evaluation metrics include Absolute Trajectory Error (ATE) for localisation, Chamfer Distance (CD) for geometric fidelity, and rendering indicators such as PSNR, SSIM, and LPIPS [3].

B. 3D Gaussian Splatting

3DGS represents scenes as collections of 3D Gaussians, enabling real-time novel-view synthesis at rates exceeding 100 fps—and up to 200 fps in optimised configurations [11]. In urban-scale scenes, 3DGS achieves PSNR values of up to 28.93 dB, SSIM scores of 0.918, and LPIPS as low as 0.056, outperforming NeRF baselines [3], [12]. Liu et al. [2] trace the

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evolution from classical multi-view geometry through NeRF to 3DGS.

Beyond the high rendering throughput, a critical challenge in infrastructure inspection is the drastic difference in scale and resolution between global aerial views (drone) and close-range terrestrial views (street-level or detailed inspection), which generates conflicts in geometric densification. Recent frameworks such as Horizon-GS [12] address this limitation by unifying both perspectives through a coarse-to-fine training strategy coupled with Level-of-Detail (LOD) structures, achieving consistent photorealism regardless of the camera distance and perspective.

C. Vision-Language-Action Models

VLA models extend large vision-language models by incorporating an action output head, mapping visual observations and language instructions to robot control commands. Qwen2.5-VL [4] achieves spatial reasoning accuracies between 75.6% and 86.3% on the EmbSpatial benchmark. Lumo-1 [5] integrates purposeful robotic control with multi-modal perception via chain-of-thought reasoning. SwitchVLA [6] achieves task-switching success rates above 90%.

D. Next-Best-View Planning

Pred-NBV [1] learns to predict information gain of candidate viewpoints, observing up to 25.46% more surface points than conventional planners within the same temporal budget. GenNBV [7] generalises NBV planning across object categories via a policy network, optimising the flight path length versus coverage trade-off. Zhang et al. [10] demonstrate NBV-driven exploration with 3DGS-based reconstruction for semantic indoor mapping using autonomous UAVs.

E. Edge-Optimised Inference for Robotics

III. Semantic-Guided Autonomous Mapping

A. Closed-Loop Architecture

True autonomy demands that perception and action operate synergistically. We propose a continuous five-stage cycle for active exploration:

1) **Semantics (VLA):** The model processes complex natural-language instructions (e.g., "inspect the corrosion"), filters the background, and defines the Region of Interest (ROI) [5].
2) **Localisation (ViSLAM):** A visual SLAM system (such as MCGS-SLAM [8]) provides a robust estimate of the UAV pose together with an initial sparse point cloud.
3) **Reconstruction (3DGS):** Using the estimated pose, the explicit Gaussian map is densified and optimised in real time [2].
4) **Uncertainty (Pred-NBV):** The planner evaluates the partial 3DGS map to predict the next best view, maximising information gain while minimising flight effort [1].
5) **Action (Control):** The VLA model validates the safety of the generated trajectory before issuing motor commands to the drone controller, thereby closing the loop [6].

B. Edge Computing Deployment

Table I summarises the expected performance characteristics under cloud and edge deployment scenarios, based on LoRA fine-tuning and 8-bit quantisation strategies for VLA models.

TABLE I
COMPARISON OF VLA DEPLOYMENT CONFIGURATIONS

Configuration	Method	Latency (s)	VRAM (GB)
Cloud (Traditional)	Full fine-tuning	> 2.50	40–80
Edge (Optimised)	8-bit + LoRA	0.60–0.85	8–12

The 3DGS rendering pipeline achieves frame rates above 100 fps [11], while feed-forward reconstruction variants reduce per-scene latency to 0.29–0.63 s [11]. Zhang et al. [10] have demonstrated the feasibility of 3DGS-based semantic reconstruction on autonomous UAV platforms.

C. Semantic Uncertainty Modulation

Unlike traditional geometric frontier approaches, uncertainty in a 3DGS environment encompasses both geometric and radiometric factors. This includes depth estimation uncertainty, position gradients (the magnitude of changes in the Gaussian centres μ during optimisation), and entropy along rendered rays [2]. In the proposed architecture, the VLA's semantic understanding modulates these uncertainty metrics by assigning a substantially higher probabilistic weight to zones that match the user's instruction. Consequently, the system prioritises uncertainty reduction (re-scanning) over critical elements such as pillars or cracks, while ignoring high entropy in semantically irrelevant regions such as the sky or background foliage [4].

IV. Comparative Analysis

This section synthesises quantitative performance metrics from the surveyed literature along four evaluation axes.

A. Reconstruction Quality Metrics

Key metrics from the literature:

- **PSNR:** 3DGS achieves up to 28.93 dB in urban scenes [12].
- **SSIM:** Structural similarity scores of 0.918 [3].
- **LPIPS:** Perceptual similarity as low as 0.056 [3].
- **ATE:** Absolute Trajectory Error for SLAM localisation accuracy.
- **CD:** Chamfer Distance for point cloud geometric fidelity.

B. Exploration Efficiency Metrics

Key metrics from the literature:

- **Coverage Rate / AUC:** Measures how rapidly the planner achieves full coverage.
- **Surface Point Gain:** Pred-NBV achieves 25.46% more observed points [1].
- **Flight Path Length vs. Coverage:** GenNBV optimises this trade-off across object categories [7].

C. Cognitive Performance Metrics

Key metrics from the literature:

- **Task Switching Success Rate:** SwitchVLA achieves $>$ 90% [6].
- **Spatial Reasoning Accuracy:** Qwen2.5-VL achieves 75.6%–86.3% on EmbSpatial [4]; Lumo-1 exhibits comparable performance [5].

D. Hardware Feasibility Metrics

Key metrics from the literature:

- **Rendering FPS:** 3DGS exceeds 100 fps (up to 200 fps) [11].
- **Reconstruction Latency:** Feed-forward methods achieve 0.29–0.63 s per scene [11].
- **VLA VRAM:** Cloud $\approx$ 40–80 GB $\rightarrow$ Edge $\approx$ 8–12 GB (8-bit + LoRA).
- **VLA Latency:** Cloud > 2.50 s $\rightarrow$ Edge 0.60–0.85 s.

V. Discussion

A. Latency–Accuracy Trade-offs on Edge Devices

The simultaneous deployment of VLA models and 3DGS reconstruction on-board edge devices (e.g., NVIDIA Jetson) presents a severe computational bottleneck. While cloud-based systems can accommodate massive models and unconstrained rendering, a UAV operates under strict battery and payload limits, with a typical VRAM budget of 8–12 GB. This makes parameter-efficient adaptations imperative, including LoRA and 8-bit quantisation for VLA models. By applying these optimisation techniques, inference latency can be reduced to sub-second levels (0.60–0.85 s), maintaining a viable balance between the accuracy of semantic actions and the response speed required for safe flight [10].

B. Dynamic Environments and Scene Changes

Operational infrastructures are dynamic environments where inspection priorities may change abruptly. Traditional VLA models assume a static intent from start to finish, causing oscillations or failures if the operator modifies the instruction mid-task. Recent advances such as SwitchVLA [6] demonstrate that it is possible to endow the system with reactive *task-switching* capabilities. By modelling task changes as a behaviour-modulation problem conditioned on the execution state, the VLA can reverse, advance, or switch tasks fluidly and autonomously with success rates exceeding 90%, without relying on external planners or manual scripts.

C. Safety-Critical Certification

D. Transferability to Unstructured Environments

E. Benchmark Standardisation

F. Multi-Robot Cooperative Exploration

The inspection of massive infrastructures demands scalability, achievable through cooperative deployment of UAV swarms. When adapting predictive planning to multiple platforms, a key challenge is the redundant overlap of observed areas. Emerging frameworks such as MAP-NBV (Multi-Agent Prediction-guided Next-Best-View) [13] address this problem by sharing partial observations to build a joint team representation. Simulations demonstrate that, by simultaneously optimising joint information gain and control effort, the collaborative method observes on average 22.75% more unique surface points in its first iteration compared with single-agent predictive approaches.

G. Long-Horizon Mission Planning

In prolonged, long-horizon mapping missions, traditional decision-making systems are susceptible to state-error accumulation and stagnation in repetitive loops (e.g., scanning the same area repeatedly). To mitigate this problem, advanced VLA models such as Lumo-1 [5] have integrated chain-of-thought reasoning modules that explicitly predict "sub-task completeness" before generating a new command. By introducing this short-term history and context, the system reliably discriminates already-inspected areas, maintaining exploration consistency over extended time windows.

VI. Conclusion

This survey has reviewed the convergence of 3D Gaussian Splatting, Vision-Language-Action models, and predictive Next-Best-View planning toward a unified framework for semantic-guided autonomous mapping. The quantitative evidence consolidated from the literature demonstrates that the integration of VLA models optimised via LoRA with geometric systems such as 3DGS is not only computationally feasible on edge hardware (VRAM $\approx$ 8–12 GB, latency < 1 s) but imperative for the next generation of mobile robots and UAVs in autonomous inspection missions.

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References

[1] H. Dhami, V. D. Sharma, and P. Tokekar, "Pred-NBV: Prediction-guided next-best-view planning for 3D object reconstruction," *arXiv preprint arXiv:2304.11465*, 2023.

[2] S. Liu, M. Yang, T. Xing, and R. Yang, "A survey of 3D reconstruction: The evolution from multi-view geometry to NeRF and 3DGS," *Sensors*, vol. 25, no. 5748, 2025.

[3] M. E. Atik, "Comparative assessment of neural radiance fields and 3D Gaussian Splatting for point cloud generation from UAV imagery," *Sensors*, vol. 25, no. 10, p. 2995, 2025.

[4] S. Bai *et al.*, "Qwen3-VL technical report: The most powerful vision-language model in the Qwen model family," *arXiv preprint arXiv:2511.21631*, 2025.

[5] Astribot Team, "Mind to hand: Purposeful robotic control via embodied reasoning (Lumo-1)," *arXiv preprint arXiv:2512.08580*, 2025.

[6] M. Li, Z. Zhao, Z. Che, F. Liao, K. Wu, Z. Xu, P. Ren, Z. Jin, N. Liu, and J. Tang, "SwitchVLA: Execution-aware task switching for vision-language-action models," *arXiv preprint*, 2025, beijing Innovation Center of Humanoid Robotics.

[7] X. Chen, Q. Li, T. Wang, T. Xue, and J. Pang, "GenNBV: Generalizable next-best-view policy for active 3D reconstruction," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, shanghai AI Laboratory / CUHK.

[8] Z. Cao, H. Wu, L. W. Tang, Z. Luo, Z. Zhu, W. Zhang, M. Pollefeys, and M. R. Oswald, "MCGS-SLAM: A multi-camera SLAM framework using Gaussian Splatting for high-fidelity mapping," *arXiv preprint*, 2024.

[9] B. Lee, J. Park, K. T. Giang, S. Jo, and S. Song, "MVS-GS: High-quality 3D Gaussian Splatting mapping via online multi-view stereo," *IEEE Access*, vol. 13, pp. 1–13, 2025.

[10] H. X. Zhang, Y. Yang, and Z. Zou, "Autonomous unmanned aerial vehicles exploration for semantic indoor reconstruction using 3D Gaussian Splatting," *Data-Centric Engineering*, vol. 6, p. e38, 2025.

[11] X. Li *et al.*, "DrivingForward: Feed-forward 3D Gaussian Splatting for driving scene reconstruction from flexible surround-view input," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025.

[12] T. Lu, M. Yu, L. Xu, Y. Xiangli, L. Wang, D. Lin, and B. Dai, "Horizon-GS: Unified 3D Gaussian Splatting for large-scale aerial-to-ground scenes," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

[13] H. Dhami, V. D. Sharma, and P. Tokekar, "MAP-NBV: Multi-agent prediction-guided next-best-view planning for active 3D object reconstruction," *arXiv preprint*, 2024, university of Maryland.